USN | | | | | | | | | | [22EC33]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

MICROCONTROLLER AND ITS APPLICATIONS

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.
ii. Answer All Questions

		PART-A		**5 X 2=10**	
1.	**a.**	State which of the following is unidirectional and bidirectional. **i.** Data bus **ii.** Address bus	L2	C01	**02**
	b.	List any 2 features of PIC18F452 microcontroller.	L1	C02	**02**
	c.	Write an ALP to perform addition of two 8- bit literal data using PIC18F458 microcontroller.	L3	C03	**02**
	d.	What is Working Register of a PIC18F452 microcontroller?	L1	C04	**02**
	e.	What do you mean by Serial Baud Rate Generator (SBRG)?	L1	C06	**02**
		PART-B		**6 X 5=30**	
2.	**a.**	Find the total width of the address bus and data bus for an 8-bit CPU and 8-bit instruction set architecture. The size of program memory is 512K bytes, and the size of data memory is 2K bytes.	L3	C01	**05**
		OR			
	b.	Distinguish between Microprocessor and Microcontroller architectures.	L2	C01	**05**
3.	**a.**	Write a PIC18F452 assembly program to transfer/copy 8-bit Literal data-32H to 5 memory locations of File Registers (40H, 41H, 42H, 43H and 44H).	L3	C02	**05**
		OR			
	b.	Illustrate the PIC18F458 status register. Explain each field.	L2	C02	**05**
4.	**a.**	Discuss the various ports of a PIC18F458/2 microcontroller.	L2	C03	**05**
		OR			
	b.	What are advantages of C language over assembly level language?	L2	C03	**05**
5.	**a.**	With relevant diagrams, describe simplex, half duplex and full duplex communications.	L2	C05	**05**
		OR			

	b.	State the advantage and disadvantage of parallel communication over serial communication.	L2	C04	**05**
6.	a.	Illustrate the programming steps of PIC18F458/2 timer0 to generate time delay in 16-bit mode.	L2	C02	**05**
		OR			
	b.	List the pins/signals used for ADC interfacing of PIC18F458/2.	L2	C02	**05**
7.	a.	Write a C18 program to convert data from BCD "0x29" to its ASCII values of PIC18F458.	L3	C02	**05**
		OR			
	b.	Write a C18 program for a PIC18F458 microcontroller to read data from RB0 and send it to RC7 after inverting it.	L3	C02	**05**
		PART-C		**6 X10=60**	
8.	a.	Explain the role of the following inside CPU: **i.** Registers **ii.** Program Counter **iii.** Instruction Decoder **iv.** ALU **v.** Bus System	L2	C01	**10**
		OR			
	b.	With neat diagram, describe the CPU's instruction execution cycle in detail.	L2	C01	**10**
9.	a.	Explain the following instructions for the PIC18F458 microcontroller: **i.** MOVLW K **ii.** ANDLW K **iii.** IORLW K **iv.** MOVWF address **v.** MOVFF address1, address2	L3	C02	**10**
		OR			
	b.	Explain the various C data types supported by the C18 compiler of the PIC18F458 microcontroller. Also give the data size and range of each data type.	L2	C03	**10**
10.	a.	A door sensor is connected to the RB1 pin, and a buzzer is connected to RC7. Write a C18 program to monitor the door sensor, and when it opens, sound the buzzer	L4	C03	**05**

	b.	Write a C18 program to get a byte of data from Port B, wait 500milli seconds, and then send it to Port C.	L4	C03	**05**
		OR			
		Write a C18 program for the following data conversions for the PIC18F458 microcontroller: **i.** ASCII data "0x39" & "0x37" values to Packed BCD value "0x97". **ii.** Hexa value "0XCB" to BCD value "203".	L4	C03	**10**
11.	a.	A switch is connected to pin RD7. Write a C program to monitor the status of switch(SW) and perform the following: **i.** If SW = 0, the stepper motor moves clockwise. **ii.** If SW = 1, the stepper motor moves counterclockwise. Draw an interface diagram for a Stepper motor connected to PORTD of a PIC18F458.	L4	C06	**10**
		OR			
	b.	Explain how to interface a relay to a PIC18F458 microcontroller using ULN2803 driver. Create a C program to control ON/OFF of the AC bulb via a relay using a microcontrolier.	L4	C06	**10**
12.	a.	Explain the role of the driver circuit for interfacing a DC motor with microcontroller. DC motor is connected to RB0 and switch which is connected to RD7 of PIC18F458 microcontroller. Also Write a C program to monitor the status of SW and control the speed of the DC motor using PWM: **i.** If SW = 0, the DC motor moves with 50% duty cycle. **ii.** If SW = 1, the DC motor moves with 25% duty cycle.	L4	C06	**10**
		OR			
	b.	Draw a 16x2 LCD pin diagram and describe each pin in detail. Also give commands to configure the 16X2 LCD.	L2	C06	**10**
13.	a.	Write a C program to interface the 7 Segment display to a PORTD of the PIC18F458 microcontroller. And also display 0 to 9 on 7 segment display with some delay.	L4	C05	**10**
		OR			
	b.	Explain how 4×4 keypad is interfaced with PIC18F458 microcontroller and write a C program for keypad scanning.	L3	C06	**10**